

Reason2Act: Aligning Reasoning and Actions in Vision–Language–Action Models with Verifiable Rewards

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Abstract—Vision–language–action (VLA) models increasingly generate intermediate embodied reasoning before producing robot actions, with the goal of improving generalization and interpretability. However, we observe that the stated reasoning of such policies can be inconsistent with the actions they subsequently execute: a policy may, for example, state that the end effector should move toward an object while generating an action chunk that moves away from it. We study this *reasoning–action misalignment* and introduce a set of physically verifiable consistency measures that compare motion, target, and subtask intent expressed in reasoning with the resulting robot action trajectory. We then use these signals as rewards for reinforcement learning, directly optimizing a VLA for reasoning–action consistency without human preference labels or a learned reward model. Experiments in [TODO: simulation benchmark(s)] show that REASON2ACT improves reasoning–action alignment by [TODO: XX%], task success by [TODO: XX%], and out-of-distribution performance by [TODO: XX%] over [TODO: baseline(s)]. These results suggest that physically grounding the relationship between what a robot says and what it does provides a simple and scalable objective for post-training reasoning-capable VLAs.

I. INTRODUCTION

Vision–language–action (VLA) models have emerged as a promising paradigm for general-purpose robot control, leveraging large-scale vision–language pretraining together with diverse robot demonstration data to map multimodal observations and language instructions to robot actions [1]–[9]. Beyond direct action prediction, Recent work further augments VLAs with embodied chain-of-thought (ECoT) reasoning that explicitly describes task plans, spatial relations, motions, and intermediate subgoals before acting [10], [11].

However, an important failure mode remains underexplored: *the actions executed by a VLA policy do not necessarily align with the intermediate reasoning it produces*. In preliminary experiments with MiniVLA on LIBERO-90, we observe that the model identifies a locally appropriate motion but subsequently executes an action in a conflicting direction. Moreover, these reasoning–action inconsistencies occur more frequently in failed episodes than in successful ones, suggesting a meaningful relationship between reasoning–action alignment and downstream task performance. We provide a detailed quantitative analysis of this phenomenon in the experiments. Figure 1 illustrates two representative failures.

This mismatch is distinct from whether a reasoning trace is linguistically plausible or whether an episode eventually succeeds. Statements such as move left, move forward and up, or move down imply concrete physical behavior that can be verified directly against the subsequent action trajectory. We therefore represent each stated motion as a three-dimensional

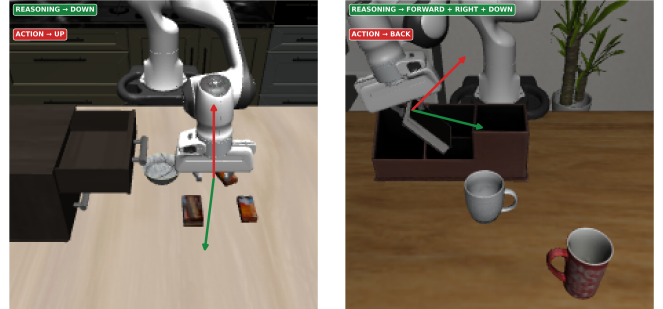


Fig. 1: Left (pick and place butter): the reasoning proposes Move Down to reach and grasp the butter, but the executed action moves up. Right (book placement): the reasoning proposes Move Forward, Right, and Down to align the book with the front compartment, but the executed action moves backward. Green arrows show reasoning-implied motion; red arrows show executed end-effector motion.

Move vector and compare it with the accumulated end-effector translation over the predicted action chunk. This yields an interpretable, physically grounded consistency verifier without relying on a learned semantic or LLM/VLM judge.

We ask: *Can physically verifiable reasoning–action consistency improve both the faithfulness and task performance of a reasoning-capable VLA?* We first characterize reasoning–action misalignment in [TODO: benchmark and model] and study how frequently it occurs and how it relates to downstream task performance. We then introduce REASON2ACT, a XX post-training objective that augments the task reward with automatically computed reasoning–action consistency rewards. Our contributions are:

- **Diagnosis.** We identify and quantitatively characterize reasoning–action misalignment in reasoning-capable VLA policies across [TODO: motion direction, target-relative motion, subtask and other failure modes].
- **Verification.** We formulate trajectory-based consistency metrics that directly test whether executed robot actions agree with the physical intent by the policy’s reasoning.
- **Post-training.** We introduce XXX that uses verifiable rewards to improve reasoning–action alignment without requiring human preference labels or a separately trained semantic reward model.
- **Evaluation.** We evaluate the proposed approach on [TODO: simulated manipulation tasks / benchmark suites] and study whether improved reasoning–action alignment translates to improved task performance.

II. RELATED WORK

A. Vision–Language–Action Models

Vision–language–action (VLA) models extend pretrained vision–language representations to robotic control by grounding visual observations and language instructions in robot actions. Early approaches such as RT-2 cast robot actions as tokens within a vision–language model, enabling knowledge acquired from large-scale web data to transfer to robotic manipulation [1]. OpenVLA subsequently demonstrated an open-source VLA pretrained on large robot datasets [2], while Octo explored generalist policy pretraining across heterogeneous robot embodiments and action spaces [3].

More recent work has explored alternative architectures and action-generation mechanisms. RDT-1B uses diffusion for bimanual action generation [6], while CogACT couples multimodal understanding with a specialized action module [7]. The π_0 family uses flow matching to generate continuous action chunks [4], [5]. More recent variants extend this framework with reinforcement learning for policy fine-tuning in $\pi_{0.6}$ [12] and world-model-based reasoning for improved instruction following in $\pi_{0.7}$ [13]. Recent systems such as GR00T and SmolVLA further investigate scalable generalist policies and efficient VLA deployment [8], [9].

Our goal is not to introduce a new VLA architecture, but to study the alignment between embodied reasoning and executed actions. For controlled comparison, we follow the original ECoT-Lite setup and use its MiniVLA architecture as the policy backbone [11]. While our experiments focus on ECoT-Lite, the proposed Reason2Act framework is model-agnostic and can be applied to other VLA architectures.

B. Embodied Reasoning for Robot Control

Explicit reasoning can improve VLA policies by decomposing complex instructions into task-relevant intermediate representations before action generation. [14] introduces embodied chain-of-thought (ECoT) reasoning over task plans, subtasks, grounded visual features, and robot motions, improving generalization to challenging manipulation tasks. Subsequent work extends this paradigm toward more efficient training, open-world reasoning, and tighter integration of reasoning with continuous control [11], [15]–[17]. Together, these works demonstrate the growing role of reasoning in improving performance and generalization of VLA policies.

C. Reasoning Faithfulness and Alignment

Whether explicit reasoning faithfully reflects a policy’s behavior remains an open problem in reasoning-enabled VLAs. Fast ECoT introduces *action faithfulness*, measuring how predicted actions change when intermediate reasoning steps are removed; this quantifies the dependence of action prediction on the reasoning trace, but does not determine whether the resulting behavior physically realizes the stated reasoning [18]. SEAL more directly studies reasoning–action alignment and performs training-free runtime verification: it samples candidate action sequences, predicts their outcomes, and uses a pretrained VLM to select the outcome most consistent with the policy’s textual plan [19]. Pinocchio

considers a broader notion of embodied faithfulness, using a learned critic to evaluate consistency across perception, intermediate reasoning, high-level decisions, and predicted trajectories, and subsequently uses these scores as dense rewards for reinforcement-learning post-training [20].

These approaches assess alignment through model-based semantic judgments, relying on an LLM/VLM judge [19] or a learned critic [20]. We instead observe that embodied reasoning makes claims about physical execution—what motion to perform, which target to interact with, and which subtask to accomplish—that can be directly verified against the resulting trajectory. Building on this principle, we introduce a physics-grounded evaluator that translates reasoning–action pairs into *verifiable rewards* over robot states and actions, providing a physically grounded learning signal for reasoning–execution alignment without learned semantic judgment.

III. PROBLEM FORMULATION

We consider a reasoning-capable VLA policy that, given an observation o_t and language instruction ℓ , first generates an embodied reasoning trace z_t and then predicts an action chunk of horizon H :

$$z_t \sim \pi_\theta(\cdot \mid o_t, \ell), \quad a_{t:t+H-1} \sim \pi_\theta(\cdot \mid o_t, \ell, z_t). \quad (1)$$

Following ECoT [14], the reasoning contains high-level task plan, subtasks, and embodied reasoning including robot motion, and visually grounded scene features. The action chunk is subsequently generated conditioned on this reasoning.

This architecture implicitly suggests that the action should follow the physical intent expressed in the preceding reasoning. However, conventional VLA training supervises reasoning and action prediction without explicitly enforcing their consistency. Consequently, the policy may state one intended motion while producing an action chunk that commands a different direction. We refer to this discrepancy as *reasoning–action misalignment*.

Let $\mathcal{A}(z_t, a_{t:t+H-1})$ denote whether the predicted action chunk is consistent with the action-relevant intent expressed in z_t . Our objective is to reduce the probability of misalignment,

$$\Pr_{\pi_\theta}[\mathcal{A}(z_t, a_{t:t+H-1}) = 0], \quad (2)$$

while preserving or improving downstream task performance. The specific physical intent representation, alignment criterion, and training objective used to realize $\mathcal{A}$ are introduced in Section IV.

Reasoning–action consistency is distinct from reasoning correctness. An action may faithfully follow an incorrect plan, while a correct reasoning trace may be followed by an inconsistent action. We therefore evaluate both alignment and task success: alignment measures whether the policy acts according to its stated intent, whereas task success measures whether the resulting behavior accomplishes the instruction. Specifically, we study the following three questions:

- 1) **RQ1:** How frequently does a reasoning-capable VLA produce actions that disagree with its reasoning?
- 2) **RQ2:** How is reasoning–action misalignment associated with downstream task success?

- 3) **RQ3**: Can explicit alignment-aware post-training improve both reasoning–action consistency and task performance?

IV. METHOD

A. Extracting Verifiable Motion Intent

We consider an ECoT-Lite MiniVLA policy that generates intermediate reasoning before predicting a robot action chunk. As one concrete instantiation of our general reasoning–action alignment framework, we focus on the explicit **MOVE** component. We choose Move alignment because its directional intent is physically grounded, can be deterministically verified against the cumulative Cartesian displacement commanded by the predicted action chunk, and admits a differentiable cosine-similarity objective for post-training. Although this paper studies motion agreement between reasoning and robot realization, the same framework can be instantiated with other verifiable reasoning components, such as rotation, gripper operation, target selection, and subtask completion.

We first parse the robot’s **MOVE**-related ECoT reasoning into a three-dimensional motion vector whose components correspond to the forward–backward, left–right, and up–down axes:

$$d_t^{\text{reason}} = [d_t^{\text{fb}}, d_t^{\text{lr}}, d_t^{\text{ud}}] \in \{-1, 0, 1\}^3. \quad (3)$$

Each component independently specifies the requested direction along its corresponding axis. Under our simulator coordinate convention, $d_t^{\text{fb}} = -1, 0$, and $+1$ denote forward, no forward–backward motion, and backward, respectively; $d_t^{\text{lr}} = -1, 0$, and $+1$ denote left, no lateral motion, and right; and $d_t^{\text{ud}} = -1, 0$, and $+1$ denote down, no vertical motion, and up. Motions along multiple axes are represented simultaneously; for example, “move up and right” is encoded as $[0, 1, 1]$. This deterministic extraction requires neither human annotation nor a learned semantic/VLM judge.

B. Move Alignment

The policy predicts an action chunk of length $H = 10$. We aggregate its translational components to obtain the predicted displacement

$$\hat{d}_t^{\text{act}} = \sum_{h=0}^{H-1} \hat{a}_{t+h}^{xyz}, \quad (4)$$

where $\hat{a}_{t+h}^{xyz} \in \mathbb{R}^3$ denotes the predicted Cartesian translation at action step $t + h$.

We measure directional agreement using cosine similarity:

$$c_t = \frac{\langle d_t^{\text{reason}}, \hat{d}_t^{\text{act}} \rangle}{\|d_t^{\text{reason}}\|_2 \|\hat{d}_t^{\text{act}}\|_2 + \epsilon}. \quad (5)$$

A policy query is considered Move-aligned when

$$c_t \geq 0.5, \quad (6)$$

corresponding to an angular difference of at most 60° . Otherwise, it is classified as Move-misaligned.

During evaluation, we replace the predicted displacement in Eq. (5) with the realized end-effector displacement:

$$d_t^{\text{exec}} = p_{t+H} - p_t, \quad (7)$$

Observation and instruction $\rightarrow$ ECoT-Lite MiniVLA[2pt]
 $\text{MOVE reasoning} \rightarrow d_t^{\text{reason}}$
 predicted action chunk $\rightarrow \hat{d}_t^{\text{act}}$ [2pt] $\downarrow$ cosine verifier and
 thresholded hinge loss $\downarrow$ [2pt] $\mathcal{L}_{\text{performance}} + \lambda \mathcal{L}_{\text{move}} \rightarrow$
 gradient update

Fig. 2: **Overview of Move-alignment fine-tuning.** A deterministic parser extracts the intended motion from ECoT-Lite reasoning. Its direction is compared with the cumulative translation predicted by the action chunk, producing a physically interpretable alignment loss.

where $p_t \in \mathbb{R}^3$ is the Cartesian end-effector position immediately before executing the first action a_t , and p_{t+H} is its position immediately after executing the final action a_{t+H-1} in the H -step action chunk.

C. Motion Alignment Aware Fine-Tuning

A standard cosine loss, $1 - c_t$, continues rewarding examples that already satisfy the alignment criterion. This can improve average cosine similarity without necessarily reducing the fraction of queries below the 0.5 threshold. We therefore introduce a normalized thresholded squared-hinge loss:

$$\mathcal{L}_{\text{move}} = \frac{1}{N} \sum_{i=1}^N \left[\frac{\max(0, 0.5 - c_i)}{1.5} \right]^2, \quad (8)$$

where N is the number of applicable Move queries in a training batch. The normalization factor 1.5 is the maximum possible deficit from the threshold because cosine similarity is bounded below by -1 .

Equation (8) assigns zero loss to aligned queries and increasingly penalizes queries farther below the alignment threshold. In particular, its gradient magnitude increases with the deficit $0.5 - c_i$, placing greater emphasis on strongly contradictory motions.

We combine this objective with the original MiniVLA action-prediction loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{performance}} + \lambda \mathcal{L}_{\text{move}}, \quad (9)$$

where λ controls the strength of Move-alignment supervision. The alignment term is computed directly from differentiable predicted actions, so no reinforcement-learning algorithm or learned reward model is required.

V. EXPERIMENTAL SETUP

A. Policy and Benchmark

We evaluate the pretrained ECoT-Lite MiniVLA policy on LIBERO-90, comprising 90 language-conditioned manipulation tasks. The policy produces an embodied reasoning trace followed by a 10-step action chunk. Our fine-tuning set contains one rollout per LIBERO-90 task, totaling 90 episodes and 1,557 policy queries. Of these, 1,157 queries originate

from successful episodes and 400 from failed episodes. All post-training runs are initialized independently from the same untouched pretrained checkpoint. Evaluation uses two seeded trials for each of the 90 tasks, giving 180 episodes per checkpoint. Each episode contains at most 40 policy queries and terminates early upon task success. We use the standard LIBERO simulator success condition for each task.

B. Fine-Tuning Configuration

We fine-tune the policy using LoRA with rank 16 for 200 optimizer steps and an effective batch size of 32. We keep the optimizer, learning rate, scheduler, data split, training seed, and all other hyperparameters fixed across experiments.

We evaluate four alignment weights:

$$\lambda \in \{0.1, 0.3, 0.5, 1.0\}. \quad (10)$$

The current comparison includes the untouched pretrained policy and the four Move-aligned checkpoints. A matched $\lambda = 0$ fine-tuning run using the same data and optimization configuration remains necessary to isolate the effect of the alignment term from that of post-training alone.

C. Metrics

a) *Task success.*: We report

$$\text{SuccessRate} = \frac{\# \text{ successful episodes}}{\# \text{ evaluation episodes}}. \quad (11)$$

b) *Average Move cosine.*: We average the cosine similarity in Eq. (5) across all applicable policy queries.

c) *Move-misalignment rate.*: For policy query t , define

$$M_t = \mathbf{1}[c_t < 0.5]. \quad (12)$$

The overall mismatch rate is

$$\text{MismatchRate} = \frac{1}{Q} \sum_{t=1}^Q M_t, \quad (13)$$

where Q is the total number of evaluated policy queries.

d) *Outcome-conditioned mismatch.*: For episode e with Q_e policy queries, we first compute

$$M_e = \frac{1}{Q_e} \sum_{t=1}^{Q_e} M_{e,t}. \quad (14)$$

We then report the unweighted mean of M_e separately over successful and failed episodes. This prevents longer episodes from dominating the outcome-conditioned metric.

VI. RESULTS

A. Move Misalignment is Associated with Task Failure

The untouched pretrained policy succeeds in 156 of 180 evaluation episodes, corresponding to a success rate of 86.7%. Across 3,256 policy queries, its average Move cosine is 0.682, while 624 queries fall below the alignment threshold, giving an overall mismatch rate of 19.2%.

More importantly, Move misalignment differs substantially by episode outcome. Successful episodes have an average per-episode mismatch rate of 12.2%, whereas failed episodes have

an average mismatch rate of 32.4%. Thus, failed episodes exhibit approximately $2.7\times$ as much Move misalignment as successful episodes, with a gap of 20.2 percentage points. Although this association does not establish causality, it motivates testing whether directly optimizing Move consistency can improve downstream control.

TABLE I: **Move misalignment in the untouched pretrained policy.** Outcome-conditioned values are unweighted averages of per-episode mismatch rates.

Episode outcome	Episodes	Move mismatch (%)
Successful	156	12.2
Failed	24	32.4

B. Fine-Tuning Improves Alignment and Success

Table II summarizes the fine-tuning results. All four post-trained checkpoints improve average Move cosine relative to the untouched policy. However, the effect on the thresholded mismatch rate and task success depends on λ .

The $\lambda = 0.5$ checkpoint provides the strongest alignment result. It increases average cosine from 0.682 to 0.716 and reduces overall Move mismatch from 19.2% to 17.6%. It also improves task success from 86.7% to 88.3%, corresponding to three additional successful episodes.

The $\lambda = 1.0$ checkpoint achieves the highest task success, solving 162 of 180 episodes (90.0%). Its average cosine reaches 0.717, although its overall mismatch rate is 18.2%, slightly higher than that of $\lambda = 0.5$. These results suggest that average cosine, thresholded mismatch, and task success capture related but non-identical properties.

TABLE II: **Move-alignment fine-tuning on LIBERO-90.** Success is measured over 180 episodes. Overall mismatch is pooled over policy queries. Successful- and failed-episode mismatch values are unweighted means of per-episode rates.

Checkpoint	Task success	Avg. cosine $\uparrow$	Overall mismatch $\downarrow$
Original	156/180 (86.7%)	0.682	624/3256 (19.2%)
$\lambda = 0.1$	154/180 (85.6%)	0.696	607/3300 (18.4%)
$\lambda = 0.3$	153/180 (85.0%)	0.695	612/3283 (18.6%)
$\lambda = 0.5$	159/180 (88.3%)	0.716	565/3214 (17.6%)
$\lambda = 1.0$	162/180 (90.0%)	0.717	558/3070 (18.2%)

C. Effect of the Alignment Weight

Increasing the alignment weight produces a consistent overall shift toward greater reasoning–action agreement. Average Move cosine increases from 0.682 for the original policy to 0.716 and 0.717 at $\lambda = 0.5$ and 1.0, respectively. All fine-tuned policies reduce the mismatch rate relative to the original 19.2%. Although the rate increases slightly to 18.2% at $\lambda = 1.0$, the absolute number of mismatched queries decreases further, from 565 to 558. This ratio change occurs because $\lambda = 1.0$ completes more episodes successfully and terminates earlier, reducing the total number of evaluated queries from 3,214 to 3,070.

Task-level improvements emerge at the stronger alignment weights. $\lambda = 0.5$ increases success from 86.7% to 88.3%, and $\lambda = 1.0$ further improves it to 90.0%. This result supports our central hypothesis: sufficiently strong alignment supervision reduces directional contradictions between reasoning and action, allowing the policy to execute its stated motion intent more reliably and thereby improving task completion. Overall, $\lambda = 1.0$ achieves both the highest task success and the fewest mismatched queries in absolute count, while $\lambda = 0.5$ yields the lowest normalized mismatch rate.

D. Qualitative Analysis

Figure 3 presents four successful rollouts from the post-trained policy with $\lambda = 1.0$. Panels (a) and (b) revisit the two failures of the pretrained MiniVLA model highlighted in Figure 1. In the pick and place butter example, the post-trained policy moves downward as specified by its reasoning with cos similarity score 0.979, addressing the contradiction observed during grasping in the teaser. Similarly, in the book-placement example, the policy’s forward-and-down action agrees with its stated motion intent, unlike the backward action shown in the corresponding teaser example. Panels (c) and (d) provide additional examples of aligned object transport, where the policy’s actions closely follow its stated directions while successfully completing the episodes. These examples demonstrate that post-training increases agreement between the directional intent expressed in the reasoning and the Cartesian action produced by the policy. In the moka-pot and soup-can tasks, the corresponding pretrained rollouts fail under the same evaluation conditions, whereas the $\lambda = 1.0$ policy succeeds.

VII. DISCUSSION AND LIMITATIONS

Our current method verifies only explicit translational Move statements. It does not evaluate rotational commands, gripper operations, target selection, subtask completion, or long-horizon plan correctness. Cosine similarity also measures direction but ignores displacement magnitude: a very small movement may receive a high score if its direction agrees with the reasoning.

Reasoning–action alignment is not sufficient for task correctness. If the generated reasoning specifies an incorrect motion, an action can receive a high alignment score by faithfully executing that incorrect intention. Because the current objective does not contain direct supervision for reasoning correctness, it should be interpreted as improving action consistency conditional on the generated Move intent, rather than correcting the reasoning itself.

The current results are preliminary and use a single training configuration and two evaluation trials per LIBERO task. In addition, comparison with the untouched checkpoint does not isolate alignment fine-tuning from generic additional training. A matched $\lambda = 0$ control, repeated training seeds, and additional evaluation trials are needed before making statistical claims.

Finally, outcome-conditioned mismatch rates compare different sets of successful and failed episodes across check-

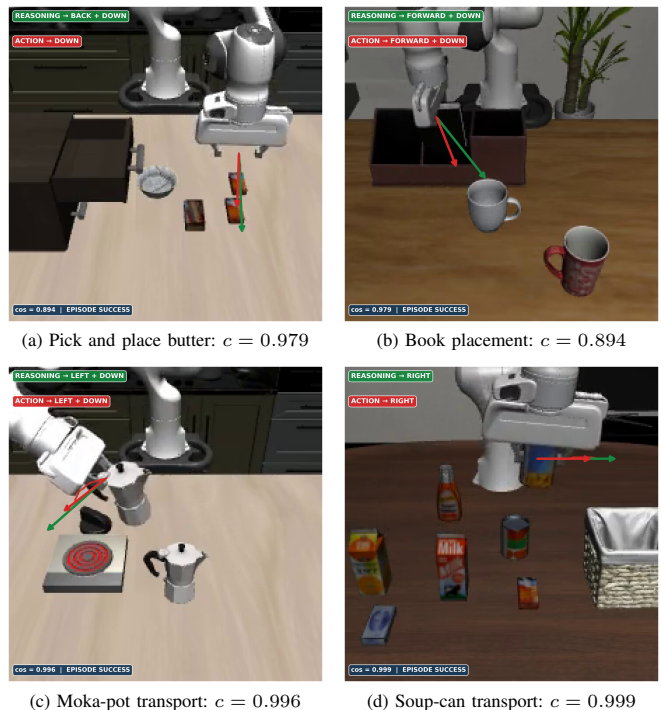


Fig. 3: **Aligned behavior after Move-alignment post-training.** Qualitative examples from the $\lambda = 1.0$ policy. Panels (a) and (b) revisit the pick-and-place and placement failure modes shown in Figure 1, now exhibiting agreement between reasoning and action. Panels (c) and (d) show additional alignment behaviors; their corresponding pretrained rollouts fail under the same evaluation conditions. Green arrows denote the reasoning-implied direction, and red arrows denote the action direction used by the evaluator.

points. Future evaluation should include paired task-and-seed transitions and confidence intervals. Extending the verifier to gripper timing, target-relative motion, and subtask completion is also an important direction, but these components are outside the scope of the present Move-only implementation.

VIII. CONCLUSION

We study a simple but consequential failure mode in reasoning-capable VLA policies: a robot can verbalize one physical intention and execute another. By using the resulting trajectory to verify motion, target, and gripper consistency, we obtain an automatic reward signal for directly post-training the coupling between reasoning and action. **[TODO: Insert final numerical results.]** Our results suggest that execution-grounded verification is a promising route toward VLA policies whose intermediate reasoning is not only interpretable, but behaviorally connected to what the robot actually does.

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